Requirements do not necessarily apply to CH 4 and/or N 2O abatement device or system. The installation requirements of the device should be respected and provided in the CH 4 and/or N2O file.
5.6.1 Permissible deviations of instruments for engine-related

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SOURCE: MEPC.402(83).pdf
abatement device or system. The installation requirements of the device should be respected and provided in the CH 4 and/or N2O file.
5.6.1 Permissible deviations of instruments for engine-related
parameters and other essential parameters
For onboard measurements only:
Engine performance and ambient condition monitoring equipment requirements should be in accordance with the requirements of 6.4.5.1.
5.9.2 CH4 and/or N 2O should be added to the list of main exhaust components. In the case of CH 4, this means CH4 as reported by the measurement device before correction for NMC efficiency.
5.9.3.2 Exhaust gas temperature at sample probe for HC
For the measurement of CH4 and/or N2O, there are no minimum temperature requirements.
5.9.6.2 Test sequence For onboard measurements only:
At each load point of a test cycle the provisions of 6.4.6.8 should apply rather than those of 5.9.6.2.
In the case of the E3 test cycle, if the actual propeller curve differs from the E3 curve, the load point used should be set using the measured engine power.
5.11 Data evaluation for gaseous emissions
In addition to the concentrations for the species to be determined

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SOURCE: MEPC.402(83).pdf
from the E3 curve, the load point used should be set using the measured engine power.
5.11 Data evaluation for gaseous emissions
In addition to the concentrations for the species to be determined
as specified by 5.11 of NTC 2008, the concentrations of CH 4

and/or N2O are to be determined. The averaged results are to be
given in ppm.
Where the NMC methane efficiency is not 0% and/or the NMC
ethane efficiency is not 100% the CH 4 concentration to be used
in equation 18a is calculated as follows:
cNMHC = cHC (w/oCutter)·(1 Em) cHC (wCutter)
 Ee - Em

Where:
cHC (wCutter) HC concentration with sample gas
through NMC (ppmC1)
cHC (w/oCutter) HC concentration with NMC bypassed
i.e. usual HC reading (cHC) (ppmC1)
Em NMC methane efficiency appendix IV, 8.5.1
Ee NMC ethane efficiency appendix IV, 8.5.2
CH4 concentration:
cCH4 = cHC - cNMHC
5.12.5.1 Calculation of
the emission mass
flow rates

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SOURCE: MEPC.402(83).pdf
i.e. usual HC reading (cHC) (ppmC1)
Em NMC methane efficiency appendix IV, 8.5.1
Ee NMC ethane efficiency appendix IV, 8.5.2
CH4 concentration:
cCH4 = cHC - cNMHC
5.12.5.1 Calculation of
the emission mass
flow rates
For the calculation of the emission mass flow rates in 5.12.5.1,
ugas values for N2O and/or CH4 should be calculated using table 5
as extended:
Gas HC CH4 N2O
gas kg/m3 * 0.716 1.9631
e Coefficient ugas
Liquid
fuel** 1.2943 0.000479 0.000553 0.001517
Rapeseed
Methyl
Ester
1.2950 0.000536 0.000553 0.001516
Methanol 1.2610 0.001133 0.000568 0.001557
Ethanol 1.2757 0.000805 0.000561 0.001539
Natural
gas
1.2661 0.000558* 0.000565 0.001551
Propane 1.2805 0.000512 0.000559 0.001533

Butane 1.2832 0.000505 0.000558 0.001530
* In the case of HC emissions when using natural gas as the fuel,
the given ugas value of 0.000558 should be used for NMHC on the
basis of CH2.93. For total HC, the ugas of CH4 should be used.
5.12.5.2 The CH 4 and/or N 2O concentration to be entered into
equation 18a is the value from 5.11 on a wet basis.
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)

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SOURCE: MEPC.402(83).pdf
basis of CH2.93. For total HC, the ugas of CH4 should be used.
5.12.5.2 The CH 4 and/or N 2O concentration to be entered into
equation 18a is the value from 5.11 on a wet basis.
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4)
AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
Annex 7, page 7
I:\MEPC\MEPC 83-17-Add.1.docx
5.12.6 Calculation of
the specific emission
Calculate CfCH4 (g/kg fuel and g/kWh), CfN2O (g/kg fuel and g/kWh)
and Cslip-CH4 (% of the mass of the methane containing fuel used
by the engine) for each load point where emissions are
measured.
Cfgas = qmgas /fuel flow
Cslip-CH4 = (qmCH4 /fuel flow)·100
Calculate the average weighed emissions
Cfgas = _i=1...n (qmgas · WFi) / sum_i=1..n (qfuel,i · WFi)
With qfuel,i being the fuel flow at each mode point.
Cslip-CH4 = _i=1...n (qmCH4 · WFi) / 10 / sum_i=1..n (qfuel,i · WFi)
qmgas (g/h): see section 5.12.5.2 of the NTC 2008, equation 18a
Fuel flow (kg/h) as measured.
Appendix III
Section 1.1
CH4 and/or N2O are added to the list of components included in
the exhaust gas analysis system.
Figure 1

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SOURCE: MEPC.402(83).pdf
qmgas (g/h): see section 5.12.5.2 of the NTC 2008, equation 18a
Fuel flow (kg/h) as measured.
Appendix III
Section 1.1
CH4 and/or N2O are added to the list of components included in
the exhaust gas analysis system.
Figure 1
For arrangements of exhaust gas analysis measurement
systems for the measurement of CH4, refer to ISO 8178-1
section 7.4.4.
NMC and N2O analyser should be arranged, installed and
operated in accordance with the respective

manufacturer's recommendations.
Section 1.2
Analysers for CH4 and N2O: see section 3
Section 3
3.6 Methane (CH4) analysis
The reference method for CH4 should be FID + NMC.
Other principles / systems should be accepted if proven against FID+NMC with exhaust gases of the compositions to be measured.
It should be ensured that the HC / CH 4 analyser(s) to be used have duly calibrated ranges for the respective concentrations to be measured.
The NMC should have the capacity to handle the expected CH 4 and NMHC concentrations.
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4) AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1

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SOURCE: MEPC.402(83).pdf
and NMHC concentrations.
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)